Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Accelerating/Fast-Tracking Digital Health Solutions Across Borders			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio-Medical		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	*Remote and Home Health care Devices (in combination with other RWD sources/ Health Information)*	*Information Security*	*Security Technologies / Service & Application Security / Network System Security*
Concept of Proposed Technology	Digital health solutions are expanding rapidly, with the global digital health market exceeding USD 330 billion and continuing to grow at over 20% annually (Dwivedi, A. (2025). Korea is exceptionally well positioned in this landscape, with more than 90% of hospitals using electronic medical records (Lee et al., 2022), a highly digitized healthcare infrastructure, and strong government support through initiatives such as the AI Data Voucher Program and national health data platforms. These strengths create a unique opportunity for Korean SMEs to move beyond local innovation and compete successfully in global markets.			

However, both Korean and European innovators face similar challenges: how to leverage diverse, real-world data into regulatory-grade evidence without compromising privacy, intellectual property, or national sovereignty.

For South Korean SMEs, this collaboration opens access to one of the world's largest and most innovation-driven health technology markets. Europe's medical technology sector exceeds €170 billion annually (MedTech Europe, 2025) and includes more than 500,000 types of regulated devices and diagnostics (European Commission figure as cited by Fortune Business Insights, 2025) that increasingly depend on continuous real-world evidence under the EU Medical Device Regulation (MDR) (Bando et al., 2024). Demand for digital health solutions is rising rapidly, with the European digital therapeutics market projected to grow from €0.86 billion in 2024 to €4.64 billion by 2032 (Data Bridge Market Research, 2024), creating opportunities for software, AI-driven decision support, and data-validation technologies aligned with stricter evidence requirements (Bando et al., 2024; Bolislis et al., 2020). Robust real-world evidence has proven to cut early review timelines from circa 6 - 8 months to circa 1 - 2 months, with full decisions in 3 - 4 months in accelerated pathways (Grover et al., 2025), and enable label expansions in 3 - 10 months instead of 3 - 6 years for new traditional studies (Bolislis et al., 2020); registry-based follow-up can further shorten post-market studies (Dhruva et al., 2023). Collectively, MDR-aligned solutions provide a strong European quality label that facilitates expansion into the UK, Middle East, and Asia-Pacific, regions within a global market where Europe already accounts for 26.4% of revenues (MedTech Europe, 2025).

The project will develop an **end-to-end Real-World Evidence infrastructure**. This infrastructure includes Privacy-Enhancing Technologies (PETs), enabling collaboration and analysis without sharing raw patient data. In addition, it provides regulatory-ready data pipelines, which enables machine-readable validation aligned with regulatory requirements. Verifiable AI models aim for transparency, traceability, and bias control by design. To achieve this, we combine several complementary expertises. Modern zero-knowledge proof (ZKP) systems offer powerful tools for cryptographically verifying AI model training and usage. They make it possible to prove, without revealing sensitive details, that a specific dataset or training method was used, or that a particular model was applied in the correct context or domain. We provide the expertise to design and implement tailor-made ZKP solutions that enable trustworthy and compliant deployment of AI systems. In addition, we strengthen verifiability by combining data-driven AI (such as machine learning and generative models) with symbolic reasoning (neurosymbolic AI) to improve explainability and traceability of decisions. We support bias detection, especially in textual and language-based systems, to ensure fairness and robustness. Finally, through norm engineering, we translate laws and regulations into machine-readable rules, enabling automated and auditable compliance checks within AI workflows.

This will lead to shared compliance tooling, which must lower entry barriers and administrative burden for SMEs.

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Rationale for Joint R&D Project	Healthcare systems in both Korea and Europe are under increasing pressure from ageing populations, rising chronic disease prevalence, and growing demand for digital and AI-enabled care, which require other expertise and processes. At the same time, regulatory frameworks, particularly the EU Medical Device Regulation, require continuous, real-world validation of safety and performance, creating significant barriers for SMEs (Tarricone et al., 2020). Current certification processes are slow, expensive, and poorly suited to adaptive digital technologies, often forcing smaller companies to delay market entry or limit their international ambitions. Without shared infrastructure and standardized approaches to Real-World Evidence, many SMEs struggle to scale beyond pilot deployments. TNO translates complex regulatory demands into practical technical frameworks, allowing companies to meet high compliance standards while speeding up the development and certification of new digital health solutions. By building shared, regulatory-ready Real-World Evidence infrastructures, grounded in privacy-by-design, verifiable AI, and standardized validation pipelines, TNO enables faster, more affordable, and scalable certification of digital health innovations. This gives Korean SMEs the confidence to innovate, provides regulators with trustworthy and transparent evidence, and allows healthcare systems to adopt AI-enabled solutions without compromising safety, compliance, or societal values. Korea already demonstrates the feasibility of a RWE approach. The AI Data Voucher Program demonstrates that medical AI developers can generate valuable clinical insights from safely governed, processed, and de-identified clinical data, as the program explicitly supports the secure and privacy-compliant use of hospital data for AI development (The World Korea, 2025; Yonhap Infomax, 2025). TNO provides complementary technology and regulatory expertise to make this approach scalable and internationally viable (find more on our website about PETs): - Accelerated certification: Real-World Evidence pipelines designed for trust, traceability, and privacy (Bando et al., 2024); - Privacy-Enhancing Technologies (PETs): including federated learning, secure multi-party computation, and encrypted analytics, so raw data never leaves its source - Regulatory-ready data pipelines: aligned with both Korean and European frameworks such as MDR and the European Health Data Space (EHDS); - Verifiable AI: combining cryptographic proofs (ZKP), neurosymbolic AI for explainability, bias detection, and machine-readable rules for automated compliance; - SME-focused enablement: modular, reusable tools that significantly reduce cost and complexity. By combining Korea's advanced digital health ecosystem and clinical data capabilities with TNO's PET-enabled infrastructure and regulatory know-how, this collaboration:

	- Opens access to both Korean and European markets; - Accelerates innovation without compromising safety or compliance; - Strengthens the international competitiveness of Korean SMEs. The benefits for Korean SMEs include: - Faster market access via reusable, RWE-based certification evidence: Using fit-for-purpose real-world evidence can secure indication expansions in 3-10 months instead of 3-6 years for new traditional trials, materially compressing time-to-market (Bolislis et al., 2020; Bando et al., 2024). - Safe cross-border collaboration while preserving privacy, IP, and data control: Privacy-Enhancing Technologies such as federated learning, secure multi-party computation, and synthetic data enable multi-party analysis without moving raw data, protecting confidentiality while maintaining utility - Direct alignment with European regulatory frameworks and markets, without starting from scratch: Building pipelines to EU MDR evidence expectations and the upcoming EHDS lets you generate reusable evidence for a market of circa €170 billion that represents 26.4% of the global medtech market (MedTech Europe, 2025; Bando et al., 2024). - Lower compliance risk through reusable, MDR-aligned RWE pipelines: Standardized RWE processes can replace or reduce bespoke trials in defined scenarios (e.g., label extensions), decreasing ad-hoc work and variability in submissions (Bolislis et al., 2020; Bando et al., 2024). - Earlier and more predictable revenues: In accelerated pathways, initial reviews can drop from circa 6-8 months to circa 1-2 months with full decisions in circa 3–4 months, enabling earlier sales while running post-market surveillance on the same data infrastructure (Grover et al., 2025; Dhruva et al., 2023).
Key Words	1. Digital Health Solutions 2. Privacy-Enhancing Technologies (PETs) 3. Data Sovereignty 4. Verifiable AI 5. Provenance Tracking 6. Certification Acceleration 7. Real-World Evidence (RWE) 8. SME Enablement

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (TRL 4)	End of the Project	Level (TRL 7)
R&D Objectives	Our overall project goal is to enable SMEs to use our technology for generating trusted, regulatory grade evidence from real-world data, helping them accelerate certification and market access for digital health solutions. SMEs can use the technology to - Apply RWE pipelines for design, pre-market approval, and post-market surveillance - Collaborate securely with Privacy-Enhancing Technologies (PETs; no raw data exchange) - Use regulatory-ready data pipelines and verifiable AI for compliance - Reduce cost and complexity through shared compliance tools and modular components. Possible users include digital health SMEs, MedTech innovators, telehealth providers, clinical research organisations, and regulatory service organisations. Our target is to have demonstrators validated up to TRL 7 for near-market adoption.			
R&D Target Specifications	The performance of the privacy-preserving evidence pipeline will be evaluated using both technical and regulatory metrics. For technical validation, we will measure data quality improvements in completeness, provenance accuracy, and metadata harmonization compared to baseline datasets. Furthermore, we will assess privacy guarantees through PET benchmarks to confirm that no raw patient data is exposed. For regulatory readiness, we will compare time and effort for certification preparation using our pipeline versus conventional workflows, targeting at least 30% reduction. The effectiveness of Real-World Evidence (RWE) generation will be tested in two scenarios 1) using the PET-enabled, provenance-rich pipeline, 2) using traditional siloed data workflows. Metrics will include regulatory acceptance indicators showing alignment with MDR/EHDS validation criteria, evidence completeness, reusability across lifecycle stages (design, pre-market, post-market), performance of verifiable AI models measured by bias detection accuracy, provenance traceability, and auditability scores. Usability will be assessed through System Usability Scale (SUS) and SME feedback on integration efforts, ensuring high adoption potential. Demonstrators will be validated in operational environments up to TRL 7, proving readiness for real-world deployment.			

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R&D Plan	We propose a following structure as indication. We are open to discuss alternative R&D plans. Phase 1: Foundations & Prototype (6 months) Aim: establish baselines, co-define the RWE scenario, and deliver a working PET-enabled prototype (TRL 4-5). Milestones (months) • Scenario defined & study protocols (M1): jointly agree with the pilot scenario (datasets, endpoints, evaluation), governance (PIPA/GDPR/EHDS), consent, and ground-truth plan. • Requirements & architecture baseline (M2): map MDR/EHDS checks to machine-readable rules; freeze modular architecture (PET layer, data pipeline, verifiable-AI). • PET-enabled pipeline prototype(M3-4): sketch privacy enhancing data exchange protocols and workflows; audit/output checking. • Provenance & metadata harmonization (M4-5): capture of lineage/metadata; quality flags; QC reports. • Verifiable-AI v1 & feasibility (M6): first model suggestion with bias checks and audit logs; feasibility comparison vs. conventional workflow; Phase-2 plan. Phase 2: Integration, Pilots & Validation (24 months). Aim: scale to multi-site pilots, integrate interoperability, and validate end-to-end RWE in operational settings up to TRL 7. Milestones • M6: Multi-site PET pilot (≈M9): ≥3 sources/organisations running federated analytics; leakage-risk and output checks. • M7: Regulatory-ready pipelines (≈M12): machine-readable validation aligned with legislation; internal mock submission with experts. • M8: Verifiable-AI v2 & lifecycle monitoring (≈M18): continuous drift/bias surveillance; action thresholds; audit packs. • M9: Scenario pilot completed (≈M21) for either design, pre-market and/or post-market; compare against siloed baseline. • M10: Performance & usability targets (≈M22): SUS usability target met; stakeholder acceptance. • M11: TRL-7 demonstrators & exploitation (≈M24): modules/tooling ready; integration guides; governance templates; go-to-market plan.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead applied research on privacy-preserving data sharing and Real-World Evidence (RWE) generation, ensuring scientific rigor and alignment with regulatory frameworks (MDR, EHDS). • Develop and validate core technologies, including PET-enabled data pipelines, provenance capture, and verifiable AI models, up to TRL 7 in operational environments. • Design evaluation protocols and metrics for technical performance (privacy, data quality) and regulatory readiness (compliance, certification efficiency). • Coordinate scenario definition and pilot studies, engaging SMEs and clinical partners to ensure relevance and usability. • Disseminate knowledge and best practices, through publications, workshops, and integration guides to support adoption by SMEs and ecosystem partners.

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Expected R&R of Korean SME Partner	Phase 1 (6 months): Set-up and Prototype • Use case and ground truth (including clinical trials): Define the clinical use case and provide ground-truth labels. Map existing traditional clinical trial protocols, endpoints, and case report forms to candidate real-world evidence endpoints (for example: device logs, clinician notes, insurance claims, and usage telemetry). • Hybrid evidence plan: Co-create a plan that combines traditional clinical evidence with real-world evidence. Agree on target claims, data sources, success metrics, sampling windows, and governance covering privacy, commercially sensitive information, and data relevant to national security or public safety. • Data readiness and technical input: Prepare initial datasets with clear schemas, quality checks, provenance, and standardized metadata. Align formats and define integration points for the privacy-preserving pipeline, ensuring no exchange of raw patient data between organizations or countries. • Regulatory and governance input (desirable): Advise on Korean privacy law, medical device rules, and consent language. Contribute to risk assessments and output-checking rules that protect intellectual property and other sensitive assets. • Prototype testing and feedback: Validate early prototypes using synthetic or anonymized data. Provide usability feedback to ensure the solution is practical for small and medium-sized companies. Review bias reporting and traceability features for transparent artificial intelligence. Phase 2 (24 months): Pilots and Validation • Pilot implementation across multiple organizations: Run the privacy-preserving workflows in operational environments with at least three organizations. Activate real-world evidence collection streams that complement ongoing or completed traditional clinical trials. • Hybrid validation activities: Execute scenario-based tests across (parts of the) product lifecycle (design, pre-market, and post-market). Produce combined evidence packs that merge traditional trial outcomes with real-world analyses, and compare results against the current, siloed way of working. • Compliance and stewardship: Help with ensuring that pilots comply with Korean privacy law and medical device requirements. Contribute to alignment with European regulations on data reuse and medical devices. • People and resource commitment: Assign technical and clinical staff for data curation, monitoring, model updates, and assembly of evidence packs. • Continuous feedback: Provide feedback about the feasibility of our innovation and future potential for adoption.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected Outcomes • Demonstrators of PET-enabled data pipelines and verifiable AI models validated up to TRL 7 in operational environments. • Hybrid evidence generation workflows combining traditional clinical data with Real-World Evidence (RWE) for design, pre-market certification, and post-market surveillance. • Machine-readable validation frameworks aligned with regulation, enabling automated compliance checks and contribute to lifecycle monitoring. • Secure multi-site collaboration protecting personal data, commercially sensitive information, and state-critical datasets, ensuring compliance with PIPA, GDPR, and sovereignty requirements.

	• Performance gains: ≥30% reduction in certification preparation effort; improved data quality (completeness, provenance accuracy); zero raw-data transfers across organisations. • Usability confirmed via System Usability Scale (SUS) and SME adoption feedback. Impact • Accelerated market access for SMEs through hybrid certification pathways that reuse RWE alongside traditional clinical evidence. • Trust and security in cross-border collaboration, safeguarding privacy, intellectual property, and national security interests. • Global competitiveness: Korean SMEs gain entry to European markets; European partners benefit from Korea's advanced digital health ecosystem. • Regulatory innovation: Demonstrates scalable, privacy- and security-preserving workflows for MDR/EHDS compliance, setting a benchmark for future health data governance. • Societal and economic benefits: Faster deployment of safe, AI-enabled digital health solutions improves patient care and reduces healthcare costs.

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Linda Oosterheert: Consultant semantic interoperability & trustworthy AI. Linda is a specialist in semantic interoperability and trustworthy AI at TNO. She develops and governs semantic standards that enable interoperable, scalable data infrastructures across dynamic ecosystems. Linda brings specific expertise in translating regulatory and organisational requirements into machine-readable semantic models for cross-border evidence generation and compliance.

Timon Brussaard: Senior business development manager & data ecosystem strategist. With over 18 years of experience in business development, ICT innovation, and ecosystem building, Timon Brussaard leads strategic partnerships between industry, government, and research organisations. He has expertise in data ecosystems, norm development, AI-driven

innovation, and data-space strategy, specialized in federated data infrastructures and digital health. His work focuses on helping SMEs scale technologies internationally by aligning business strategy, governance, and innovation roadmaps, highly relevant for Korea-EU joint digital health development.

Sarah van Drumpt: Researcher cybersecurity, public health & Privacy-Enhancing Technologies (PETs). Sarah is a researcher at TNO's Applied Cryptography & Quantum Algorithms department specializing in privacy-enhancing technologies (PETs), secure data collaboration, and the governance of sensitive health data. She has authored multiple publications on PET adoption, real-world evidence infrastructures, and cross-border health data use. Her work focuses on enabling safe, regulatory-proof evidence generation using federated analytics, MPC, and privacy-by-design architectures, core components of the proposed TIPA Digital Health Solutions program.

Bart Eichholtz: Tech & Innovation Consultant. Bart is an experienced technology and innovation consultant supporting the development, evaluation, and deployment of advanced digital solutions. He contributes expertise in high-tech systems, systems integration, and technical validation, helping SMEs and research partners translate emerging technologies into robust, scalable implementations. His role in the TIPA collaboration focuses on technical feasibility assessment, integration pathways, and supporting real-world validation environments.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization